Q1 - No non-degenerate limit for a finite discrete X

Values of F.

$F(x) = 0$ for $x < 1$, $F(x) = \sum_{k \le \lfloor x \rfloor} p(k)$ for $1 \le x < K$, and $F(x) = 1$ for $x \ge K$. Put

$$q := 1 - p(K) < 1 \quad (\text{since } p(K) > 0).$$

For $y < K$ we have $\lfloor y \rfloor \le K - 1$, so $F(y) \le \sum_{k=1}^{K-1} p(k) = q$. Hence for every y,

$$F(y) \le q \text{ if } y < K, \quad F(y) = 1 \text{ if } y \ge K.$$

The limit can only be 0 or 1.

Suppose $c_n > 0$, d_n and a cdf H satisfy $F(c_n x + d_n)^n \to H(x)$ at every continuity point x of H. Fix such an x and let $a_n = F(c_n x + d_n)^n$. By the above,

$$a_n \in [0, q^n] \cup \{1\} \quad \text{for every } n.$$

Suppose $0 < H(x) < 1$. Then for all large n, $a_n \in \left(\frac{1}{2}H(x), \frac{1}{2}(1 + H(x))\right)$. This interval excludes 1, so $a_n \le q^n$ for all large n. But $q^n \to 0$, so $a_n \to 0 \ne H(x)$, a contradiction.

Therefore $H(x) \in \{0, 1\}$ at every continuity pt point of H.

H is degenerate.

H is non-decreasing, so it has at most countably many jumps and its continuity points are dense. For any x, choose continuity points $y_k \downarrow x$. By right-continuity $H(x) = \lim_k H(y_k)$, a limit of values in $\{0, 1\}$, so $H(x) \in \{0, 1\}$ for all x.

Since $H(x) \to 0$ as $x \to -\infty$ and $H(x) \to 1$ as $x \to \infty$, the set $\{x : H(x) = 1\}$ is non-empty and bounded below. Let $x_0 = \inf\{x : H(x) = 1\}$. Because H is non-decreasing, $H(x) = 1$ for $x > x_0$ and $H(x) = 0$ for $x < x_0$, and right-continuity gives $H(x_0) = 1$. So

$$H(x) = 1_{x \geq x_0},$$

which is degenerate. So there are no $c_n > 0$, d_n for which $F(c_n x + d_n)^n$ converges to a non-degenerate cdf. □

(Intuitively, F jumps by $p(K)$ at its right endpoint $x_F = K$, so $P[M_n = K] = 1 - q^n \to 1$: the maximum is eventually equal to K with probability 1, and no affine rescaling can spread it out.)

Q2 (a) - $X \sim U(0, 1)$

$F(x) = x$ on $[0, 1]$, so $x_F = 1$. For $0 < a < 1$ and $0 \leq x \leq 1 - a$,

$$F_u(x) = \frac{F(u+x) - F(u)}{1 - F(u)} = \frac{(u+x) - u}{1 - u} = \frac{x}{1-u}.$$

With $\xi = -1$ the GP cdf is

$$G_{-1,\beta}(x) = 1 - \left(1 - \frac{x}{\beta}\right)^1 = \frac{x}{\beta}, \quad 0 \le x \le \beta,$$

(for $\xi < 0$ the support is $[0, -\beta/\xi] = [0, \beta]$). Take

$$\boxed{\beta(u) = 1 - u.}$$

Then $G_{-1,\beta(u)}(x) = x/(1-u) = F_u(x)$ for all $0 \le x \le 1-u$, so

$$\sup_{0 \le x \le 1-u} |F_u(x) - G_{-1,\beta(u)}(x)| = 0 \quad \text{for every } u \in (0,1),$$

and the limit as $u \uparrow 1$ is 0. (The excess over u is exactly uniform on $[0, 1-u]$.)

Q2 (b) — $X \sim N(0,1)$, $\beta(u) = 1/u$

Write $\bar{\Phi}(y) = 1 - \Phi(y) = \frac{1}{\sqrt{2\pi}} \int_y^\infty e^{-s^2/2} ds$. For $u > 0$, $x \ge 0$,

$$F_u(x) = \frac{\Phi(u+x) - \Phi(u)}{\bar{\Phi}(u)} = 1 - R_u(x), \quad R_u(x) := \frac{\bar{\Phi}(u+x)}{\bar{\Phi}(u)},$$

and with $\xi = 0$, $\beta = 1/u$: $G_{0,1/u}(x) = 1 - e^{-ux}$. Therefore

$$|F_u(x) - G_{0,1/u}(x)| = |e^{-ux} - R_u(x)|.$$

Reduction to $x = \delta$.

Both e^{-ux} and $R_u(x)$ lie in $[0,1]$ and are decreasing in x.

For $a, b \geq 0$, $|a-b| \leq \max(a,b)$. So for $x \geq \delta$,

$$|e^{-ux} - R_u(x)| \leq \max\left|e^{-ux}, R_u(x)\right| \leq \max\left(e^{-u\delta}, R_u(\delta)\right|),$$

hence

$$\sup_{\delta \leq x < \infty} |F_u(x) - G_{0,1/u}(x)| \leq \max\left|e^{-u\delta}, R_u(\delta)\right|).$$

Bound on $R_u(\delta)$.

Use the upper bound of the hint on the numerator (at $u+\delta$) and the lower bound on the denominator (at u); the factors $1/\sqrt{2\pi}$ cancel:

$$R_u(\delta) \leq \frac{\frac{1}{u+\delta} e^{-(u+\delta)^2/2}}{\frac{u}{1+u^2} e^{-u^2/2}} = \frac{1+u^2}{u(u+\delta)} e^{-u\delta - \delta^2/2} \leq \frac{1+u^2}{u^2} e^{-u\delta}.$$

Limit.

As $u \uparrow \infty$, $(1+u^2)/u^2 \to 1$ and $e^{-u\delta} \to 0$ because $\delta > 0$. So both $e^{-u\delta} \to 0$ and $R_u(\delta) \to 0$, and

$$\lim_{u \uparrow \infty} \sup_{\delta \leq x < \infty} |F_u(x) - G_{0,\beta(u)}(x)| = 0. \quad \square$$